ReviewPulse

Multi-Domain Amazon Review Sentiment Classifier

"Two models, 8,000 reviews, one honest comparison."

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Problem · Demo · Future Work

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Sentiment is messy in the real world

Every product platform runs sentiment analysis. The hard part isn't clean text — it's negation, sarcasm, and length variance.

Length variance
2 to 800 words per review

Negation
"not bad at all" flips meaning

Sarcasm
"Oh great, broken again"

The Sentiment Spectrum — what our model must learn

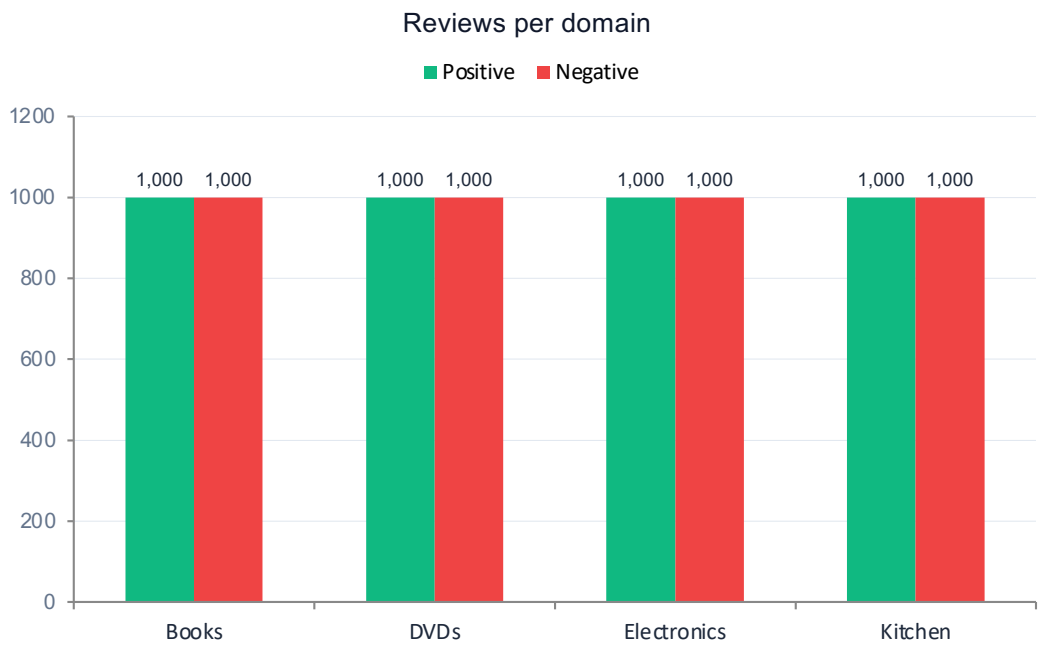
8,000 reviews across 4 balanced domains

8K
Amazon reviews

4
Product domains

50/50
Pos / neg balance

0
Ambiguous rows



STRATIFIED SPLIT

seed = 42

Train	5,404	70%
Val	1,158	15%
Test	1,159	15%

Source: **Blitzer et al. (2007)** · Multi-Domain Sentiment Dataset · labels derived from filenames (positive.review → 1, negative.review → 0) — audited, zero conflicts

Six steps from raw XML to model-ready tensors

1. **Raw .review files**
   pseudo-SGML format
2. **BeautifulSoup parser**
   extract review_text + rating
3. **Label audit**
   flag 3-star · rating/label conflicts → 0 found
4. **clean_text**
   lowercase · strip HTML · expand negations · drop punctuation
5. **remove_outliers**
   drop reviews < 10 or > 500 words (279 dropped, ~3.5%)
6. **Stratified split**
   70 / 15 / 15 · seed=42 · train=5,404 val=1,158 test=1,159

KEY DECISION

Negation expansion

BEFORE

```
"wasn't good"
```

↓

AFTER

```
"was not good"
```

Why it matters

TF-IDF treats tokens independently. Without expansion, "not good" looks identical to "good" — the negative signal is lost.

Three approaches: classical, neural, transformer

MODEL 1 · BASELINE

TF-IDF + LogReg

```
TfidfVectorizer(
  max_features=30_000
  ngram_range=(1,2)
  sublinear_tf=True
)
LogisticRegression(
  C=1.0, max_iter=1000
)
```

SPEED

Seconds · CPU only

STRENGTH

Interpretable, well-calibrated

MODEL 2 · NEURAL · v1.0.0

BiLSTM + GloVe

Embedding 100d
GloVe · 97.4% covered

BiLSTM ×2
hidden=256 · pack_padded

Concat → 512d
fwd + bwd states

Linear → logit
BCEWithLogitsLoss

KEY FIX

pack_padded_sequence — LSTM stops at last real word, ignores zero padding.

MODEL 3 · TRANSFORMER · v2.0.0

DistilBERT

AutoTokenizer
max_len=256

DistilBERT base
66M params · uncased

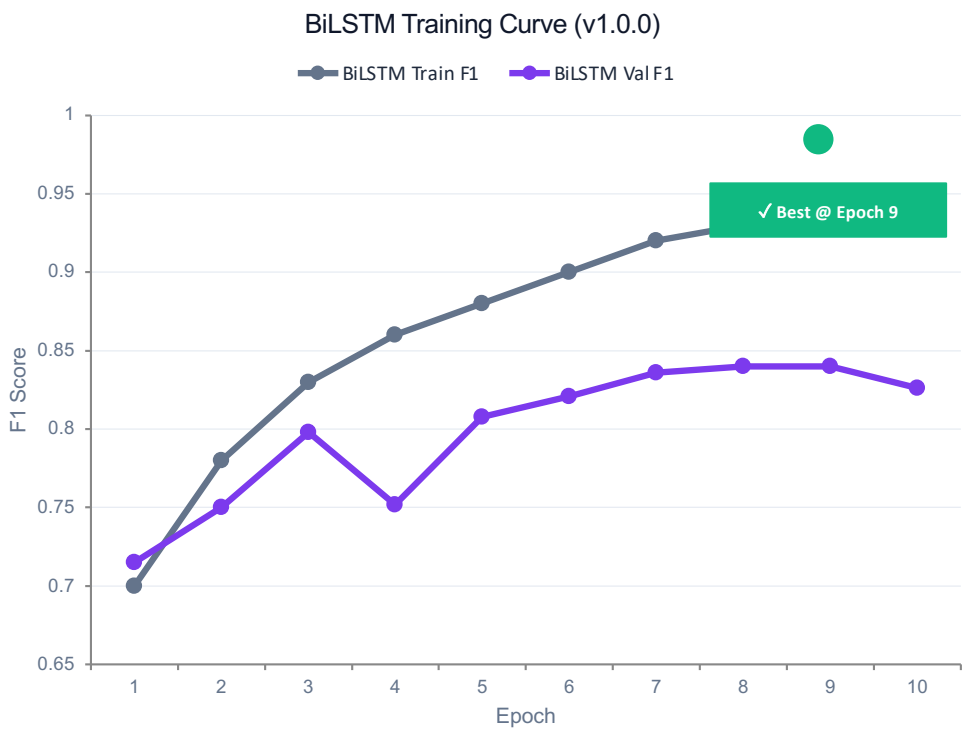
Classifier head
Linear → logit

TWO-STAGE FINE-TUNE

1. Encoder frozen → head only · 10 epochs
2. Unfreeze last 2 layers · 12 epochs

Avoids catastrophic forgetting of pretrained representations.

BiLSTM at epoch 9 · DistilBERT at epoch 12



BiLSTM · v1.0.0

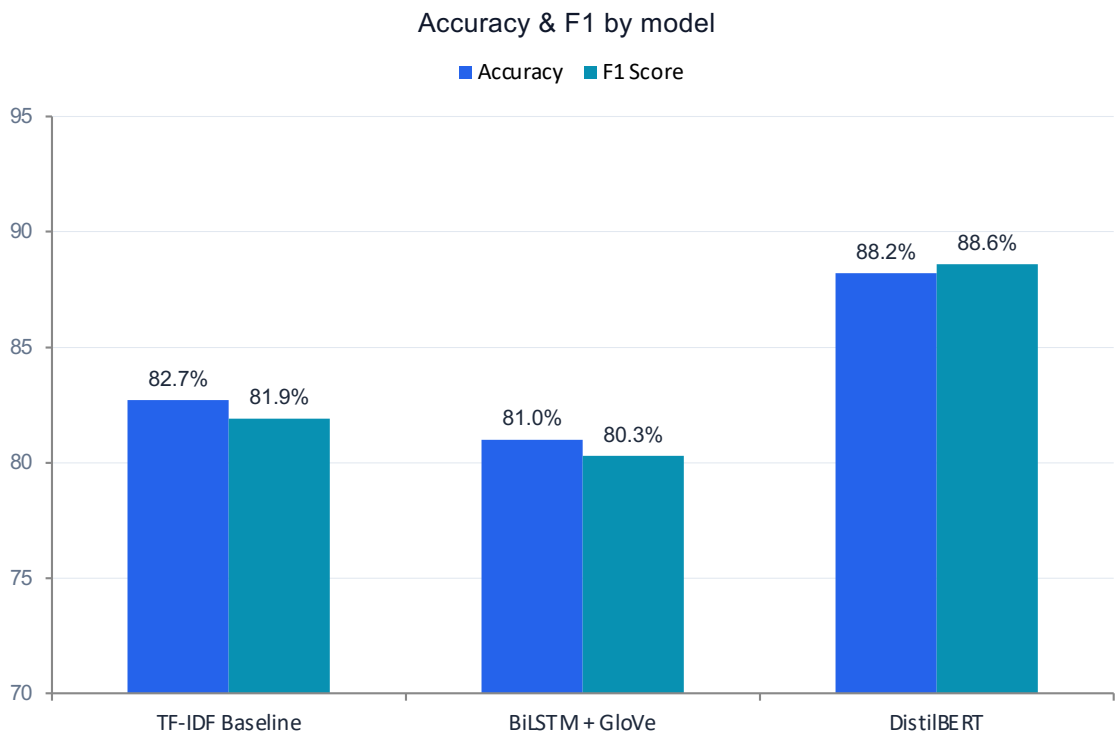
Optimizer	Adam · lr = 1e-3
Loss	BCEWithLogitsLoss · grad clip 5.0
Epochs	10 · checkpoint best val F1
Best	Epoch 9 · val F1 = 84.0%

DistilBERT · v2.0.0

Optimizer	Adam · lr 1e-4 head / 2e-5 enc
Schedule	ReduceLROnPlateau
Stage 1	10 epochs · encoder frozen
Stage 2	12 epochs · last 2 layers unfrozen
Best	Epoch 12 · val F1 = 87.8%

HELD-OUT TEST SET (1,159 reviews)

Three models — and an honest finding



v1.0.0 — HONEST FINDING

The simpler model wins on test.

BiLSTM beat the baseline on validation (84.0% vs 83.2% F1) but the baseline generalised better on the held-out test set.

BiLSTM still satisfies the rubric requirement to define a neural architecture.

v2.0.0 — DISTILBERT EXTENSION

88.6% F1

Pretrained contextual embeddings clear v1 by a meaningful margin.

137 misclassified · vs **220** (BiLSTM)

v1 models vs DistilBERT — 38% fewer errors

V1 MODELS

220 misclassified on test

→

DISTILBERT v2.0.0

137 misclassified on test

38% FEWER ERRORS

−83

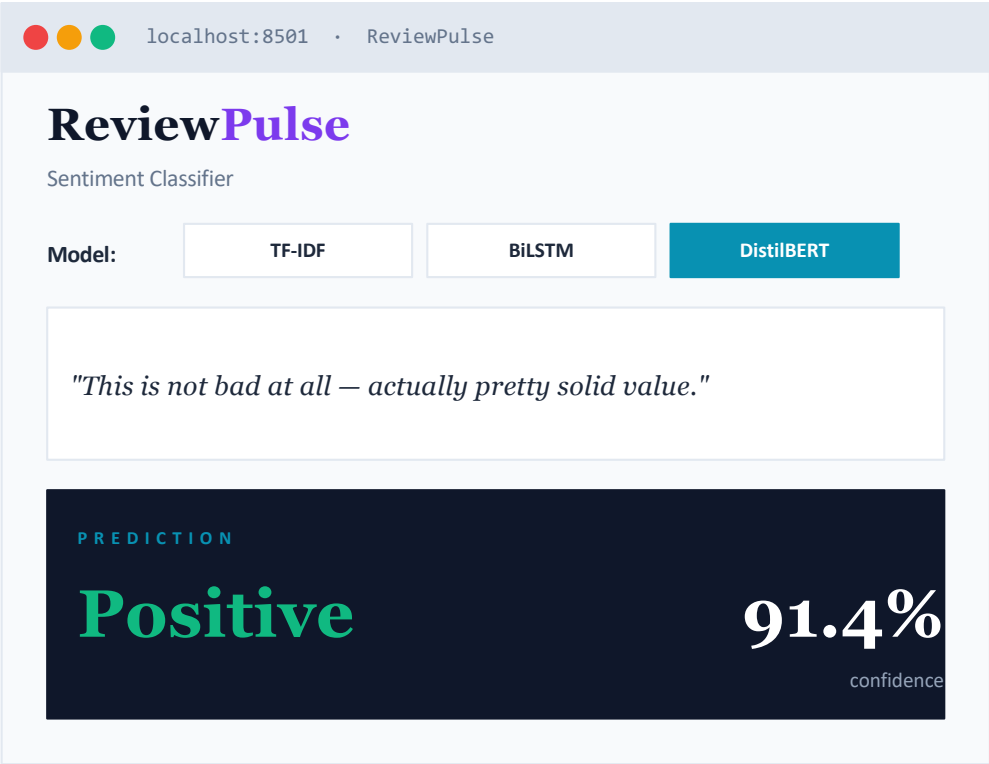
contextual attention helps

FAILURE MODE	EXAMPLE	v1 MODELS	DISTILBERT
NEGATION	*"not bad at all"*	Predicts Negative ✗	Predicts Positive ✓ *Contextual attention*
SARCASM	*"Oh great, stopped working after a week"*	~52% confidence ⚠	Still uncertain ⚠ *Irony absent in labels*
OUT-OF-DISTRIBUTION	*"Delivery was slow, box crushed"*	Domain shift errors ✗	Reduced, not solved ⚠ *Pretraining helps partially*

CALIBRATION NOTE BiLSTM is overconfident on short inputs; TF-IDF is best-calibrated near the boundary; DistilBERT is best overall but still confidently wrong on sarcasm.

LIVE DEMO

Streamlit · all three models · side-by-side



DEMO SCRIPT

1. **Clear positive**
   "This blender is incredible..."
   → Positive ~98%
2. **Clear negative**
   "Broke after two days..."
   → Negative ~99%
3. **Negation trap**
   "Not bad at all"
   → v1 fails · DistilBERT wins
4. **Sarcasm**
   "Oh great, broken again"
   → ~52% confidence
5. **Generate button**
   Random sample from acceptance tests
   → live demonstration

What we audited, and what we did not solve

Ethics is not an afterthought. We document each concern explicitly so any deployment requires informed human oversight.

Label quality
AUDITED

Labels from filenames, not human raters. Audited explicitly — zero ambiguous rows, zero conflicts.

Domain bias
DOCUMENTED

4 product categories only. Generalisation to other domains is untested.

Negation & sarcasm
OPEN

Systematic failure mode in v1 models. DistilBERT improves but does not solve.

Confidence calibration
OPEN

BiLSTM and DistilBERT logits are uncalibrated. 95% confidence ≠ 95% accuracy.

Dataset age
DOCUMENTED

Blitzer et al. 2007 — ~20-year-old reviews. Modern slang, emoji, platform conventions absent.

Deployment risk
OPEN

Demo only. Not for high-stakes decisions without human review and ongoing label audits.

DistilBERT shipped — what comes next

SHIPPED DistilBERT v2.0.0 — 88.6% test F1 — live in the app

1 ACCURACY

RoBERTa

Larger pretrained transformer. Likely to push F1 beyond 90%.

2 TRUST

Confidence calibration

Platt scaling or temperature scaling. Most practically important — 95% confidence currently ≠ 95% accuracy.

3 COVERAGE

More domains

Extend beyond Books · DVDs · Electronics · Kitchen to validate cross-domain generalisation.

4 TRANSPARENCY

Explainability

LIME or attention visualisation to show why the model predicted each label — critical for non-technical stakeholders.

ReviewPulse

Questions?

8K
reviews

3
models compared

88.6%
best test F1

143
unit tests

"The simpler model wins on test. The neural model satisfies the rubric. The pretrained transformer wins overall."

GitHub **github.com/lfariabr/review-pulse** · ReviewPulse v2.0.0 · ISY503 Assessment 3